

Selenium

Complete Installation & Setup

Beginner guide - from download to first use

Objective

By the end of this manual, you should be able to install the tool, validate the installation, perform the first functional use and recognize common errors. Do not skip validation steps: they confirm that each stage actually worked.

Before you start

Choose a language binding. This guide validates Python because it is simple for a first QA setup; modern Selenium Manager can resolve compatible drivers in common scenarios.

Official download and documentation

<https://www.selenium.dev/>

<https://www.selenium.dev/documentation/>

<https://www.selenium.dev/downloads/>

Attention

Use only test environments, accounts and devices when a tool can capture traffic, run scans, modify infrastructure or access data. In corporate environments, follow your organization policies.

Open the correct terminal

On Windows, use CMD or PowerShell. On macOS and Linux, use Terminal. Run commands exactly as shown and press Enter after each line.

Windows - step by step

Install Python and create a virtual environment in the project directory.

```
python -m venv .venv
```

```
.venv\Scripts\activate
```

```
python -m pip install selenium
```

macOS - step by step

Install Python 3 and create a virtual environment.

```
python3 -m venv .venv
```

```
source .venv/bin/activate
```

```
python -m pip install selenium
```

Linux - step by step

Install Python 3, pip and venv support.

```
python3 -m venv .venv
```

```
source .venv/bin/activate
```

```
python -m pip install selenium
```

Validate the installation

1. `python --version`
2. `python -m pip --version`

```
python --version
```

```
python -m pip --version
```

Commands and configuration

1. `python -m venv .venv`
2. `.venv\Scripts\activate`
3. `python -m pip install selenium`

```
python -m venv .venv
```

Creates an isolated virtual environment; a `.venv` directory should exist afterward.

```
.venv\Scripts\activate
```

Activates the virtual environment; the prompt normally shows `(.venv)`.

```
python -m pip install selenium
```

Installs the specified component/dependency. Wait for completion and confirm a success message.

PATH, environment variables and a new terminal

When an installer adds an executable to PATH or you configure a variable such as `JAVA_HOME/ANDROID_HOME`, close the old terminal and open a new CMD/PowerShell. Then repeat the version command. If multiple versions are installed, use `where <command>` on Windows or `which <command>` on macOS/Linux to identify the executable being used. Do not remove PATH entries without understanding their origin.

Create a safe test folder

For the first tests, use a simple folder with no production files, for example `C:\Projects\qa-lab` or `~/qa-lab`. This makes generated files, virtual environments, logs and configuration easier to identify.

First use / functional test

1. `python -c "import selenium; print(selenium.__version__)"`

```
python -c "import selenium; print(selenium.__version__)"
```

Expected result

The selenium package imports successfully and prints its installed version.

Troubleshooting

- Read the terminal error before repeating commands. Check PATH/environment variables, runtime version, permissions, system architecture and whether a new terminal was opened after environment changes.
- Command not found: confirm the installation directory is on PATH and open a new terminal.
- Wrong version: check whether another older runtime or executable appears earlier on PATH.
- Permission denied: use a user-writable project directory and follow the vendor permission guidance; do not blindly run everything as administrator/root.
- Corporate proxy/firewall: configure the package manager/tool according to your organization network policy rather than disabling security controls.

Final checklist

- Official download and documentation

- Windows - step by step / macOS - step by step / Linux - step by step
- Validate the installation
- First use / functional test
- Expected result
- Troubleshooting

Installation validated. Keep this manual for reinstallations, onboarding and troubleshooting.